

SHETH L.U.J & M V COLLEGE

SUBJECT NAME : Data Analysis with SAS / SPSS / R

PRACTICAL NO 8

AIM : Performing two-way ANOVA using aov() (R).

```
1 # Two-Way ANOVA
2
3 study_hours <- factor(c("Low","Low","High","High",
4 "Low","Low","High","High"))
5
6 gender <- factor(c("Male","Female","Male","Female",
7 "Male","Female","Male","Female"))
8
9 score <- c(60, 62, 75, 78,
10 58, 61, 73, 77)
11
12 df2 <- data.frame(study_hours, gender, score)
13
14 anova_two <- aov(score ~ study_hours * gender, data = df2)
15 summary(anova_two)
16
17
```

```
> data <- matrix(c(30, 20,
+ 15, 35),
+ nrow = 2,
+ byrow = TRUE)
> colnames(data) <- c("Product_A", "Product_B")
> rownames(data) <- c("Male", "Female")
> chisq.test(data)

Pearson's Chi-squared test with Yates' continuity correction

data: data
X-squared = 7.9192, df = 1, p-value = 0.004891
>
>
```